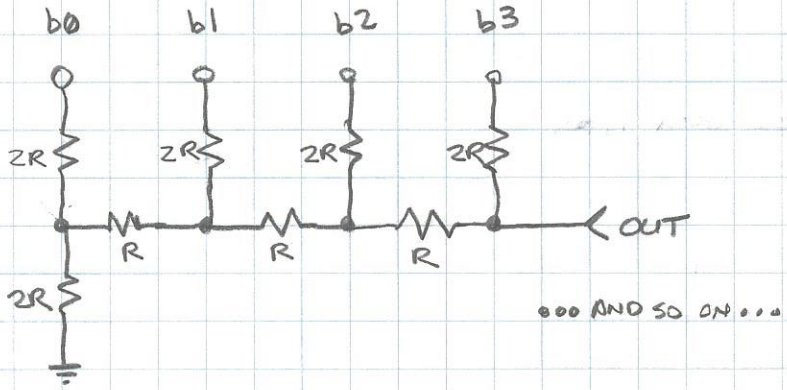
R-2R Resistor "Ladder" networks

- Used for basic Digital to Analog Conversion (DAC)



Properties

- Each bit is "binary" weighted to output
- Output resistance is R
- Only two values of resistance used (or one!)
  - ↳ makes for good accuracy

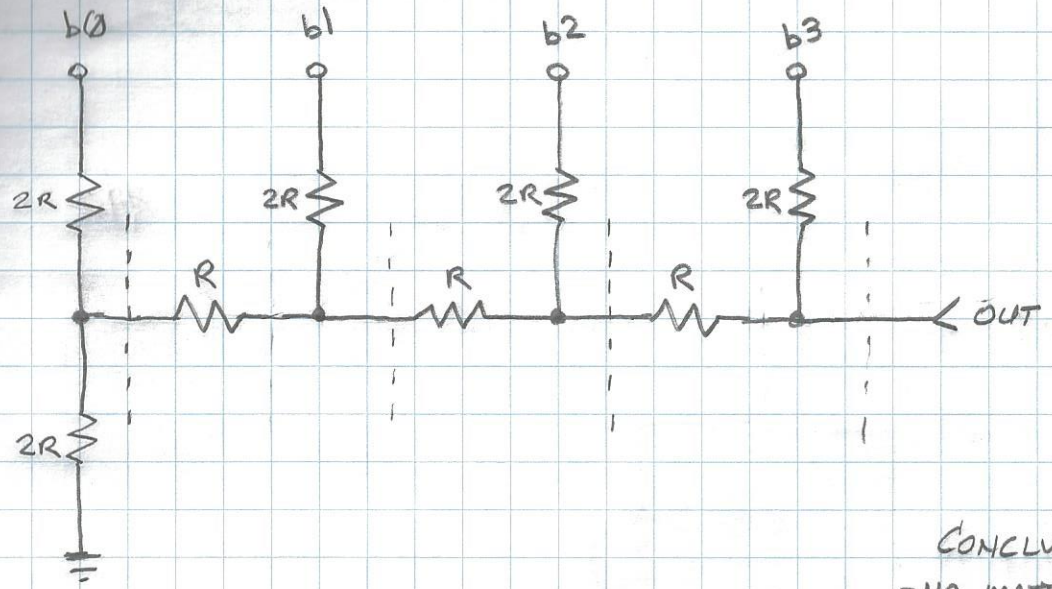
- Analysis is easy using:

1) Thevenin Equivalent Circuits

2) Superposition Principle

W2AEW

THEVENIN EQUIVALENT CIRCUITS TO CALCULATE RESISTANCE

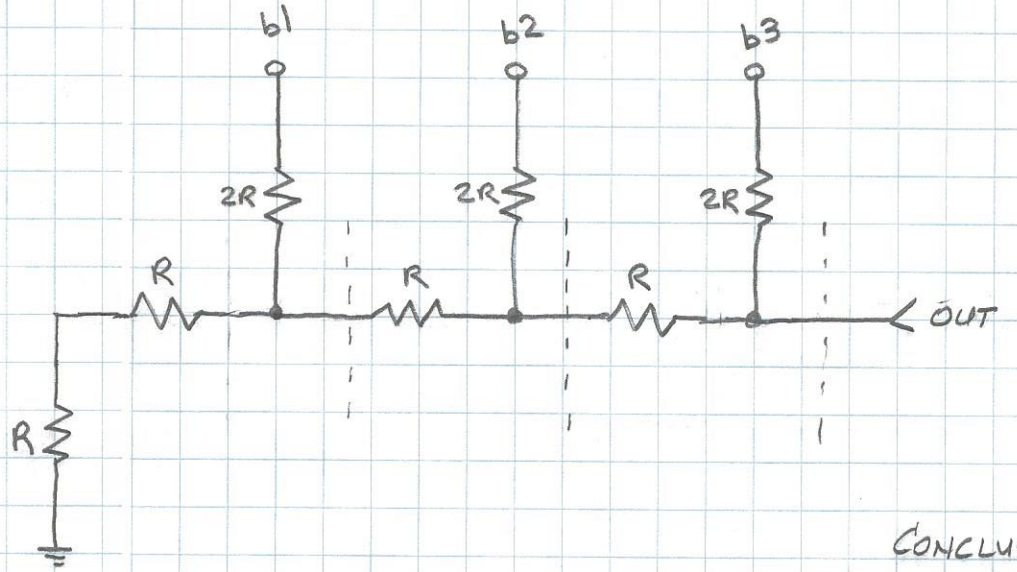


CONCLUSION –

- NO MATTER HOW MANY STAGES, OUTPUT RESISTANCE IS ALWAYS R

W2AEW

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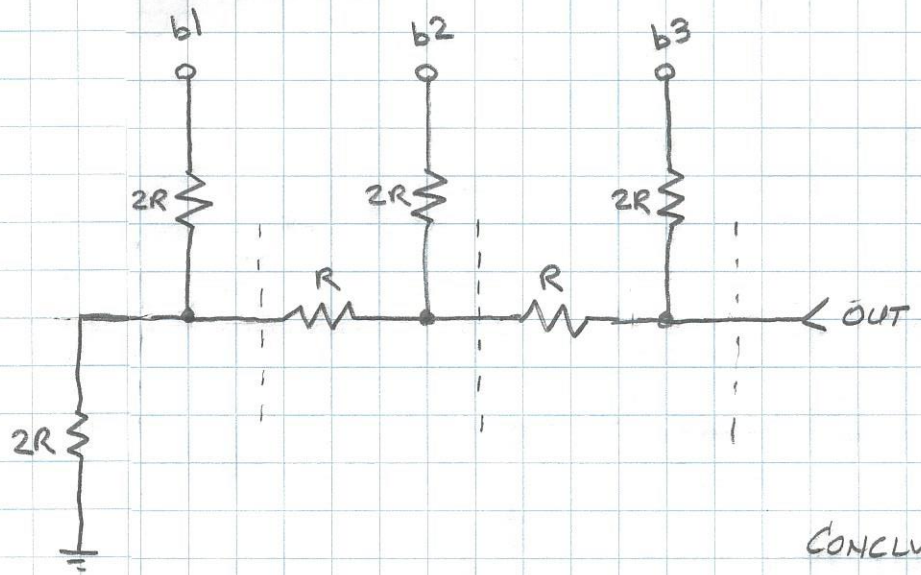


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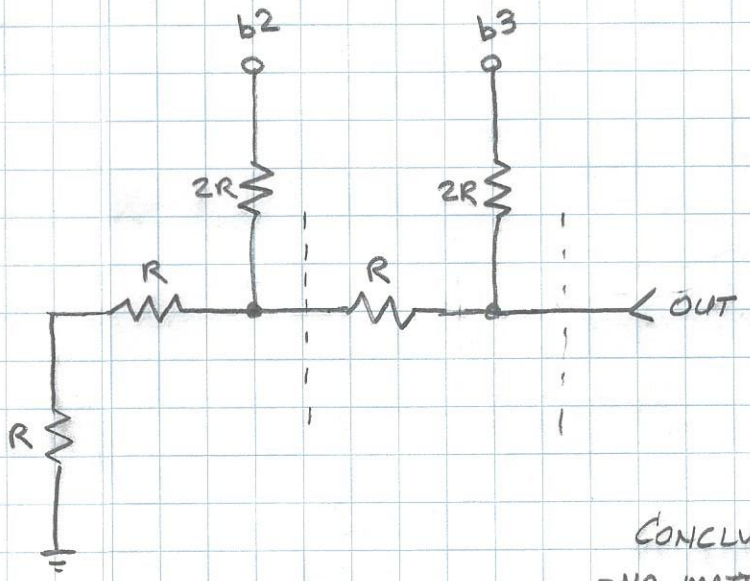
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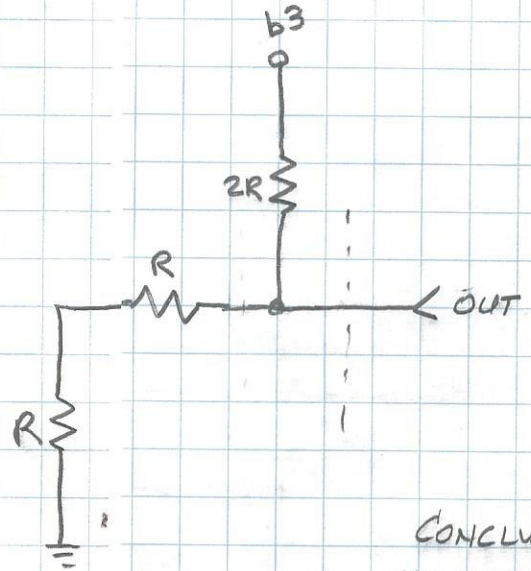
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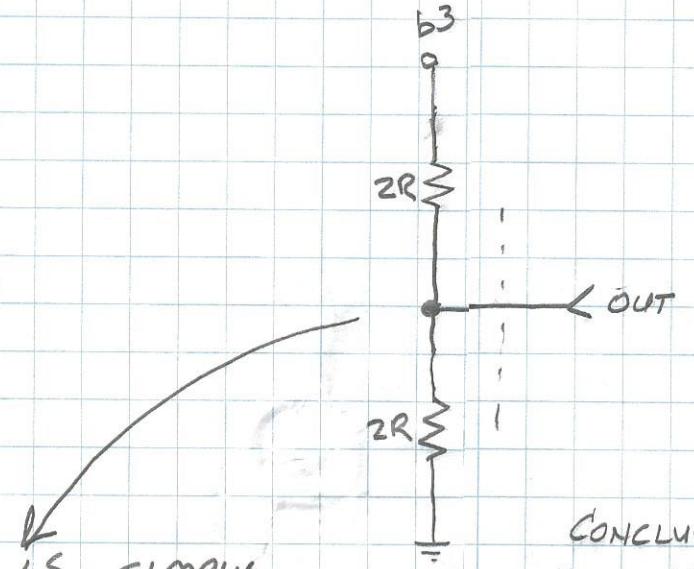


CONCLUSION –

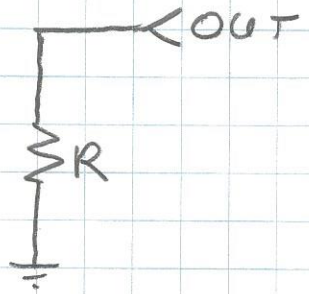
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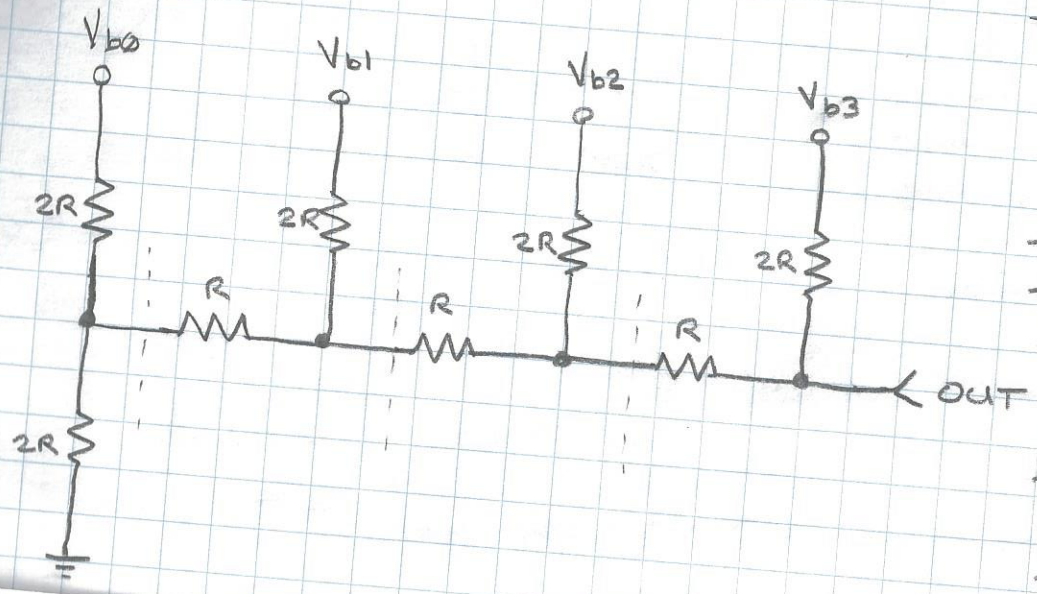
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THEVENIN & SUPERPOSITION TO CALCULATE VOLTAGE



SUPERPOSITION

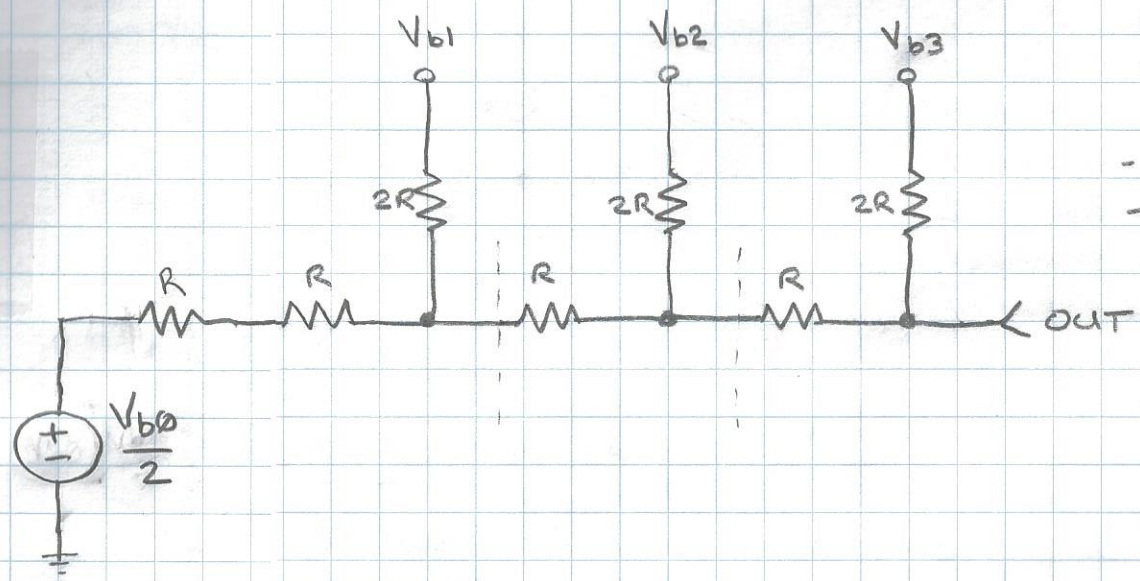
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- DO THIS FOR EACH
- SUM THE RESULTS

THEVENIN EQUIV.

- EQUIV. RESISTANCE
- OPEN CIRCUIT VOLTAGE

W2AEW

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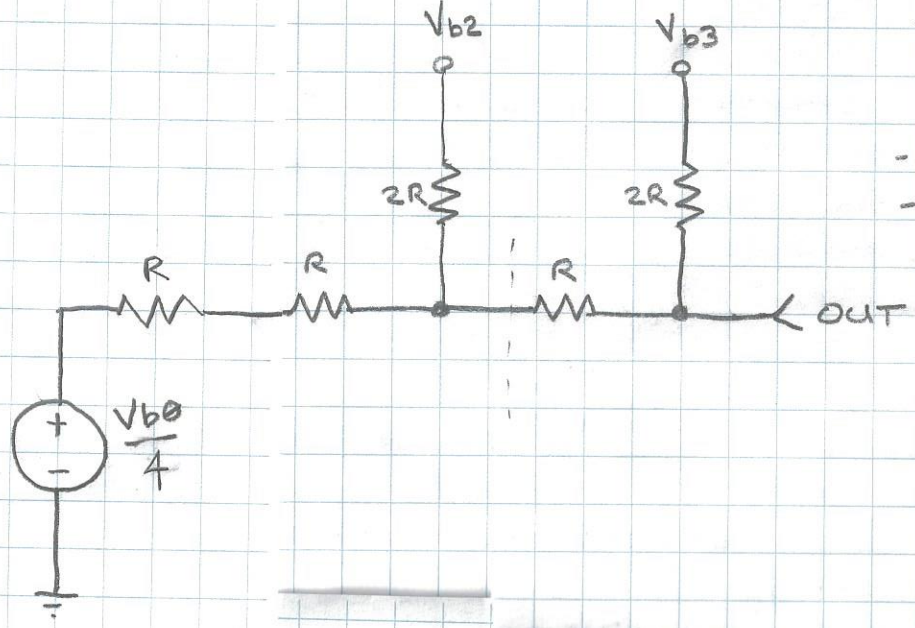
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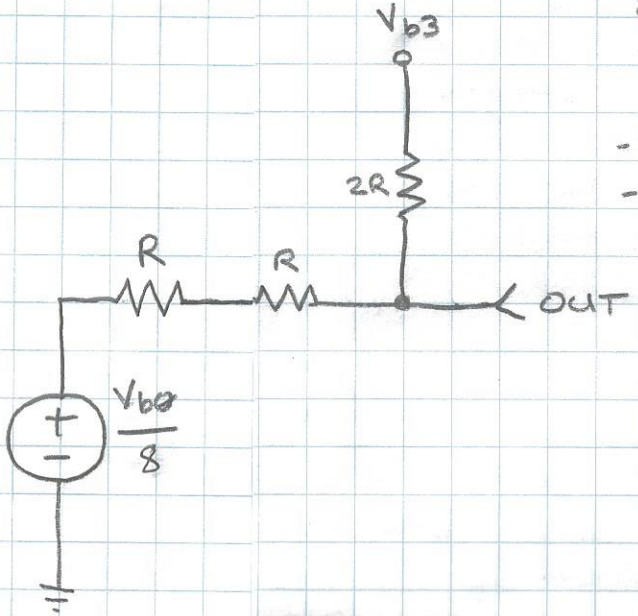
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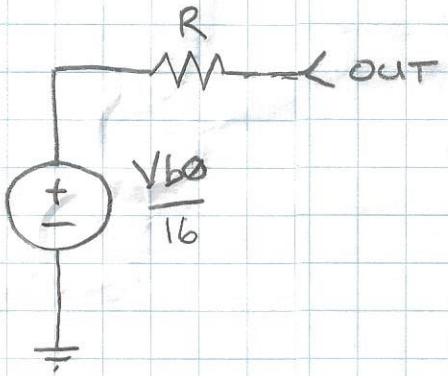
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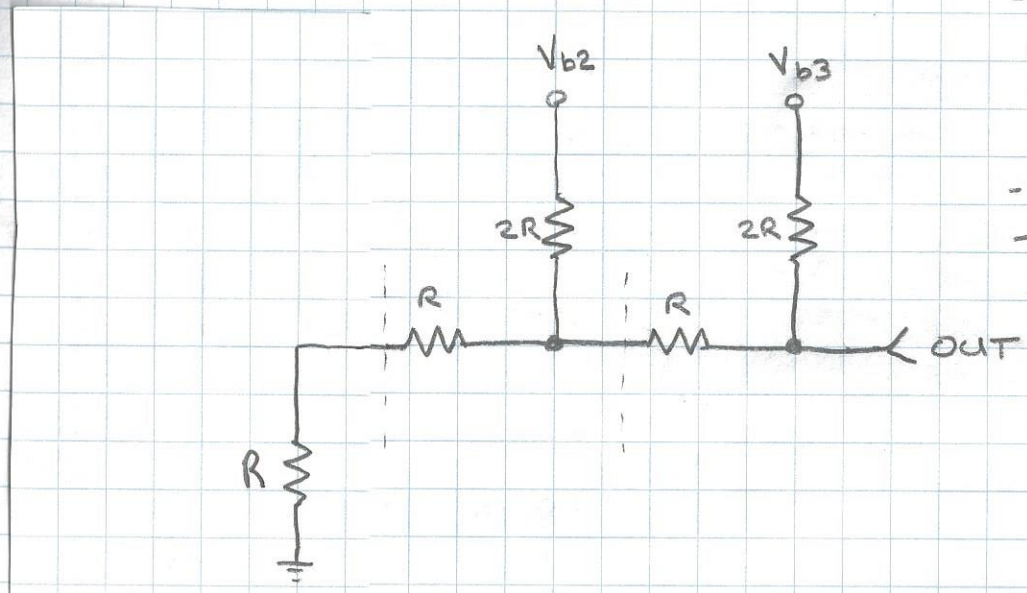
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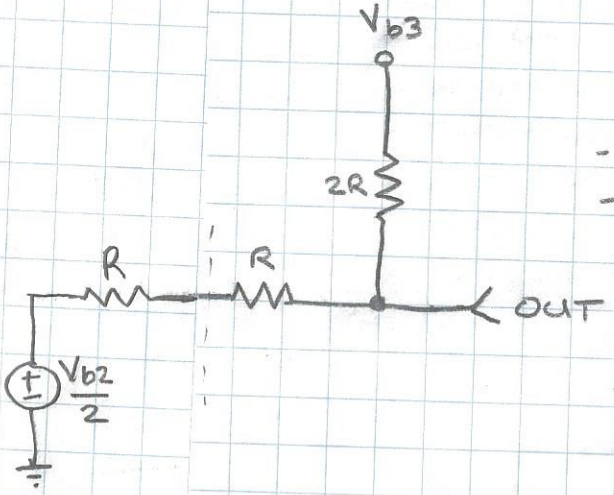
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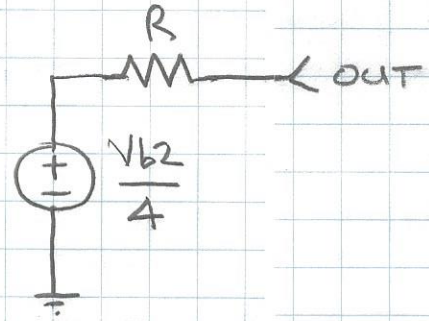
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$$V_{OUT} = \frac{V_{b0}}{16} + \frac{V_{b1}}{8} + \frac{V_{b2}}{4} + \frac{V_{b3}}{2}$$

N2AEW